* 0000800000001 * DFD

CAMBRIDGE International Education

Cambridge IGCSE™

CANDIDATE NAME

CENTRE NUMBER

CANDIDATE NUMBER

* 6 9 8 7 3 2 8 5 8 1 *

GEOGRAPHY **0460/42**

Paper 4 Alternative to Coursework **February/March 2026**

1 hour 30 minutes

You must answer on the question paper.

You will need: Insert (enclosed)
Calculator
Ruler

INSTRUCTIONS

- Answer **all** questions.
- Use a black or dark blue pen. You may use an HB pencil for any diagrams or graphs.
- Write your name, centre number and candidate number in the boxes at the top of the page.
- Write your answer to each question in the space provided.
- Do **not** use an erasable pen. Do **not** use correction fluid or tape.
- Do **not** write on any bar codes.
- If additional space is needed, you should use the lined pages at the end of this booklet; the question number or numbers must be clearly shown.

INFORMATION

- The total mark for this paper is 60.
- The number of marks for each question or part question is shown in brackets [ ].
- The insert contains additional resources referred to in the questions.

LEDCs – Less Economically Developed Countries
MEDCs – More Economically Developed Countries

This document has **16** pages.

[Turn over

1 Geography students in Wellington, New Zealand, did some fieldwork to investigate local weather. They used measuring equipment to collect data on 15 days in September. The students tested the following hypotheses:

Hypothesis 1: As atmospheric pressure increases, wind speed decreases.

Hypothesis 2: There is a relationship between daily maximum and minimum temperatures.

(a) Figure 1.1 (Insert) is a photograph of the instrument the students used to measure atmospheric pressure at 12:00 hours (midday).

(i) Name the measuring instrument shown in Figure 1.1.

.... [1]

(ii) Which unit of measurement is used to measure atmospheric pressure, shown as ‘mb’ in Figure 1.1?

.... [1]

(iii) How would the students use the moveable pointer and the measuring hand to collect atmospheric pressure data?

moveable pointer

....

....

measuring hand

....

....
[2]

(b) To measure wind speed, the students used different anemometers. Two anemometers are shown in Figure 1.2 (Insert).

(i) Choose **one** anemometer and give **one** reason why you think it is better than the other one.

Circle your chosen anemometer: **A** **B**

reason

....

.... [1]

(ii) Another student used a digital anemometer. Give **two** advantages of using a digital anemometer to measure wind speed.

1 ....

....

2 ....

.... [2]

(c) Figure 1.3 (Insert) shows a maximum-minimum thermometer.

Explain how this type of thermometer is used to measure daily temperatures.

....

....

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.... [4]

(d) The results of the students' measurements are shown in Table 1.1 (Insert).

(i) Use the results in Table 1.1 to **plot** the atmospheric pressure and wind speed recorded on 11 and 21 September on Figure 1.4. [2]

Atmospheric pressure and wind speed

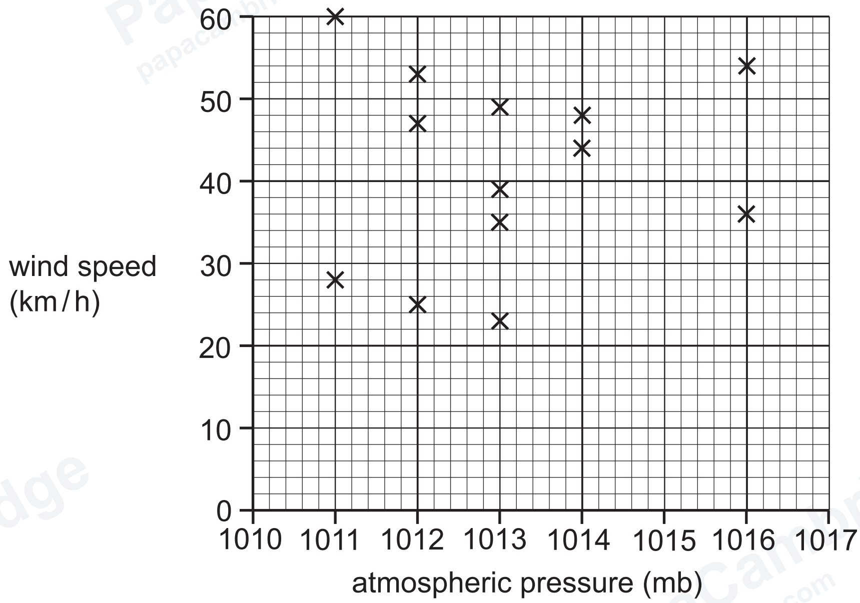

Figure 1.4

(ii) What conclusion did the students make about **Hypothesis 1**: As atmospheric pressure increases, wind speed decreases?

Support your answer with evidence from Figure 1.4 and Table 1.1.

....

....

....

....

....

.... [3]

(iii) Use the results in Table 1.1 to **plot** the maximum and minimum temperatures for 14 September on Figure 1.5. [2]

Daily temperatures

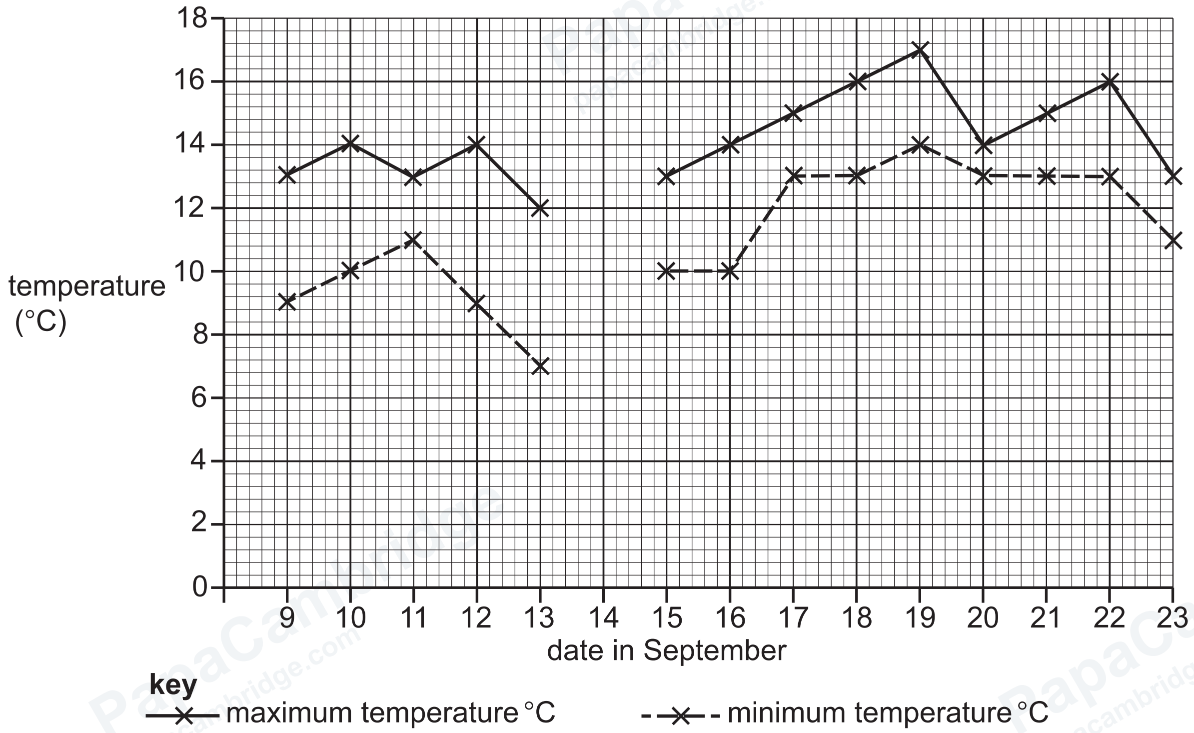

Figure 1.5

(iv) On what date in September is the smallest daily range of temperature?

.... [1]

(v) The students decided that **Hypothesis 2**: There is a relationship between daily maximum and minimum temperatures was **true**.

Do you agree with this conclusion?

Use data from Figure 1.5 and Table 1.1 to support your decision.

....

....

....

....

....

....

....

.... [4]

(e) How could the following suggestions made by the teacher make the students' fieldwork conclusions more reliable?

(i) Measure atmospheric pressure, wind speed and temperature for a month instead of 15 days.

....

.... [1]

(ii) Repeat the fieldwork by measuring atmospheric pressure, wind speed and temperature at a different time of the year before making conclusions.

....

.... [1]

(f) The students decided to extend their fieldwork by collecting rainfall data.

Suggest a hypothesis they could test, and describe a fieldwork method to help them test their hypothesis.

hypothesis to test

....

....

fieldwork method to collect rainfall data

....

....

....

....

....

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....

[5]

[Total: 30]

Turn over

2 A new supermarket was opened in the CBD of a town in the UK. Students from a local school decided to investigate the impact of this new supermarket in the town centre. Figure 2.1 (Insert) shows the location of the supermarket.

(a) **(i)** What does ‘CBD’ stand for?

C................................ B................................ D................................ [1]

(ii) Which **one** of the following is **most** likely to be in the CBD of a town?

Tick (✓) **one** answer.

	tick (✓)
airport	
bus station	
factory	
farm	
golf course	

[1]

The students investigated two hypotheses:

Hypothesis 1: The new supermarket has a positive economic impact on the town centre.

Hypothesis 2: Opening the new supermarket has made a positive impact on the environment of the town centre.

(b) To test **Hypothesis 1**, the students wrote a questionnaire to use with 100 people in the town centre. The questionnaire is shown in Figure 2.2 (Insert).

(i) Name and describe a sampling method to select people to complete the questionnaire.

name of sampling method

....

description of sampling method

....

....

....

....

.... [3]

(ii) Suggest **two** problems the students might have while using this questionnaire with people in the town centre.

1 ....

....

2 ....

.... [2]

(c) The results of the students' questionnaire are shown in Table 2.1 (Insert).

(i) Use the results in Table 2.1 to **complete** the bar for statement 2 on Figure 2.3. [2]

Questionnaire results

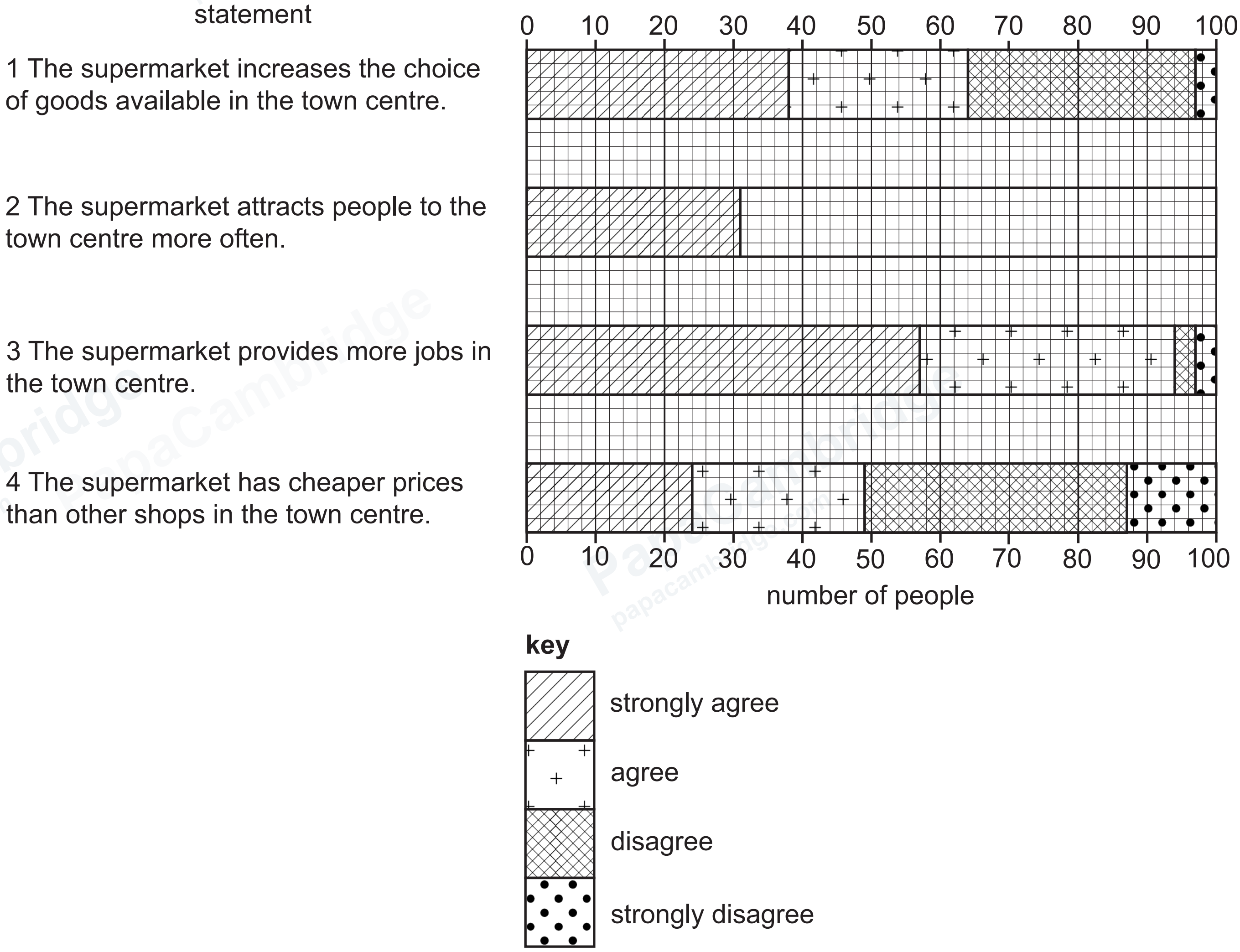

Figure 2.3

(ii) What conclusion would the students make about **Hypothesis 1**: The new supermarket has a positive economic impact on the town centre?

Use evidence from Figure 2.3 and Table 2.1 to support your answer.

....

....

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....

....

....

....

.... [4]

(d) One student devised the following formula to work out how positive the people were about the supermarket.

opinion	**points awarded**
strongly agree	4
agree	3
disagree	2
strongly disagree	1

(i) Using his formula, the student calculated the total score for each statement in the questionnaire.

Complete the following table to show the student's scores for statement 4.

Statement 4: The supermarket has cheaper prices than other shops in the town centre.

opinion	**number of answers**	**points awarded**	**score**
strongly agree	24	4	96
agree	25	3	....
disagree	38	2	....
strongly disagree	13	1	13
		total score	**260**

[1]

(ii) Figure 2.4 shows the total scores for the four statements. **Draw** the bar to show the total score for statement 4. [1]

Total scores for the four statements

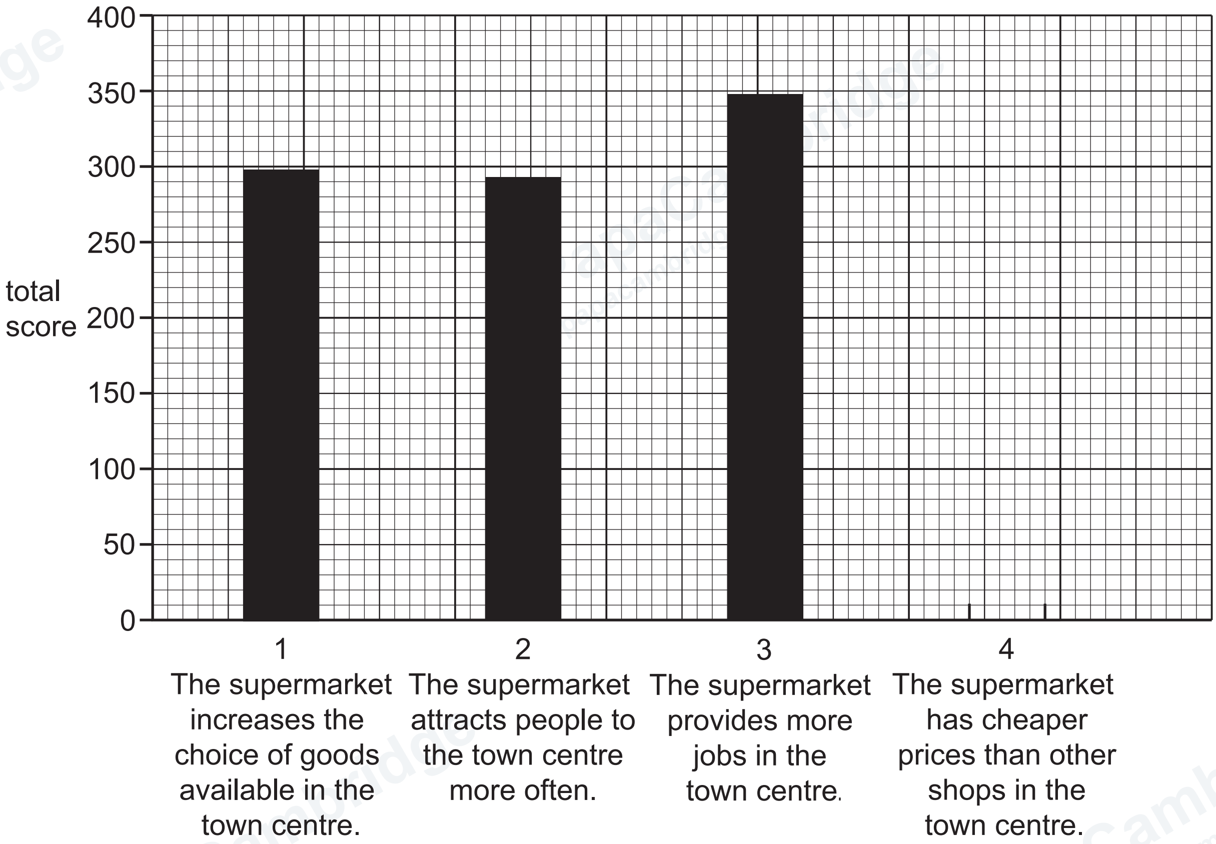

Figure 2.4

(iii) Which statement in Figure 2.4 has the most support from the people questioned?

Circle your answer.

statement 1 statement 2 statement 3 statement 4 [1]

(e) The students investigated **Hypothesis 2**: Opening the new supermarket has made a positive impact on the environment of the town centre.

They did an environmental quality survey at five locations in the town centre. The recording sheet which they used is shown in Figure 2.5 (Insert).

(i) How could the students have overcome the following difficulties of using the recording sheet?

the scoring is subjective and scores may vary between students

....

....

the score may vary at different times of day

....

.... [2]

(ii) The agreed results of the environmental quality survey are shown in Table 2.2 (Insert).

Plot the total score for location **E** on Figure 2.6. [1]

(iii) The students decided that **Hypothesis 2**: Opening the new supermarket has made a positive impact on the environment of the town centre was **true**.

Support their decision with evidence from Table 2.2 and Figure 2.6.

....

....

....

....

....

.... [3]

Results of environmental quality survey

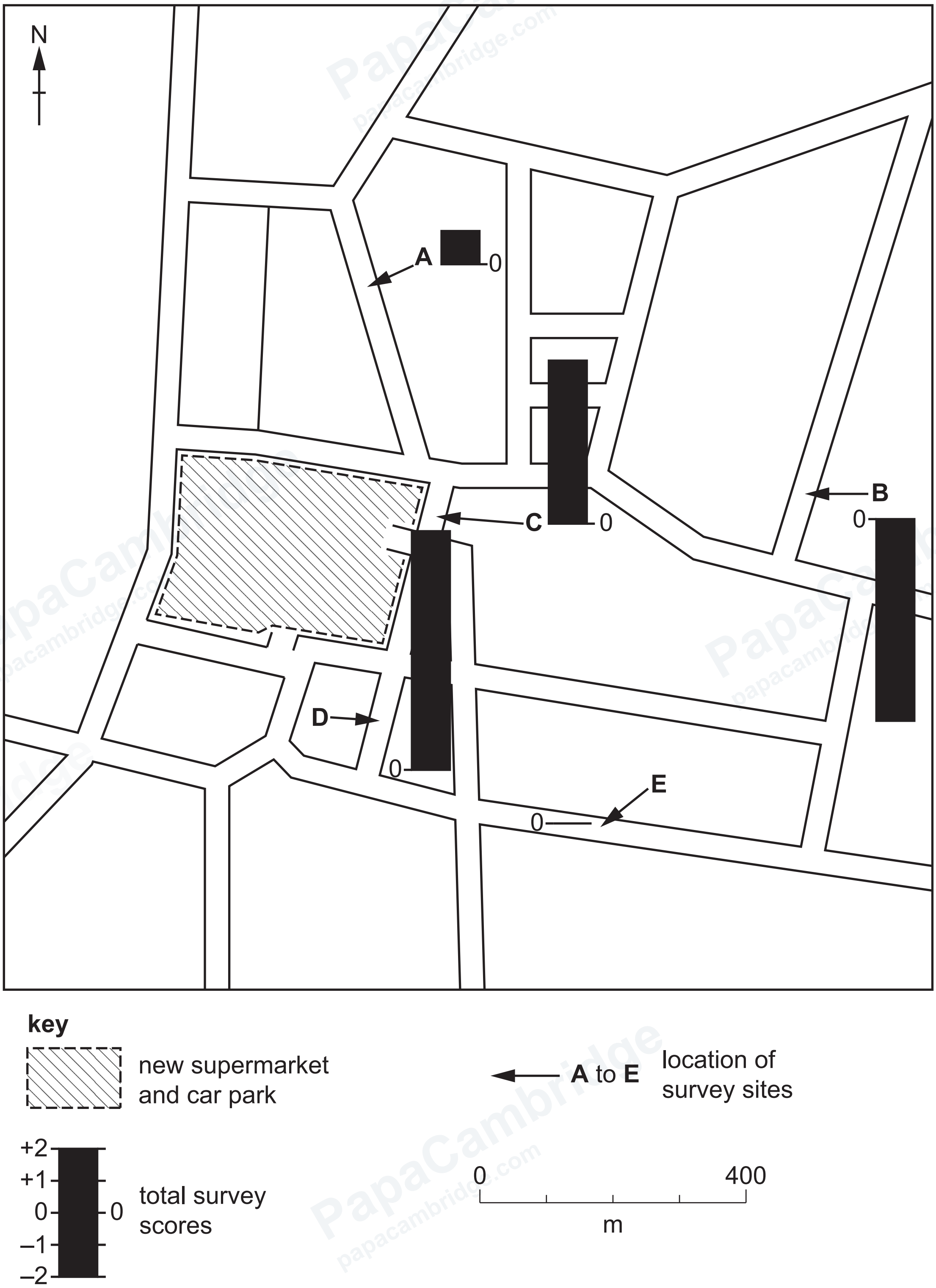

Figure 2.6

(f) To extend their investigation into one possible impact of the new supermarket, some students did a traffic count at the five survey locations (**A** to **E**) in the town centre.

The recording sheet used by the students is shown in Figure 2.7 (Insert).

(i) Which **one** of the following methods would the students use to complete their recording sheet? Tick (✓) **one** answer. [1]

method	tick (✓)
estimate	
interview	
measure	
sample	
tally	

(ii) Describe how the students would do the traffic count.

....

....

....

....

....

....

....

.... [4]

(iii) Suggest **three** difficulties the students might have when doing their traffic count.

1 ....

....

2 ....

....

3 ....

....

[3]

[Total: 30]

Additional page

If you use the following page to complete the answer to any question, the question number must be clearly shown.

Additional page

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